

Operating Systems Structure

- The operating system can be implemented with the help of various structures.
- The structure of the OS depends mainly on how the various standard components of the operating system are interconnected and merge into the kernel.
- **A system structure for an operating system is like the blueprint of how an OS is organized** and how its different parts interact with each other.

Types of Operating Systems Structures

- Simple Structure
- Layered Structure
- Modular Structure
- Virtual Machines
- Monolithic Structure
- Micro-Kernel Structure
- Hybrid-Kernel Structure
- Exo-Kernel Structure

1.Simple Structure

- Simple structure operating systems do not have well-defined structures and are **small, simple, and limited**.
- The interfaces and levels of functionality are not well separated.
- **MS-DOS** is an example of such an operating system.
- These types of operating systems cause the entire system to crash if one of the user programs fails.

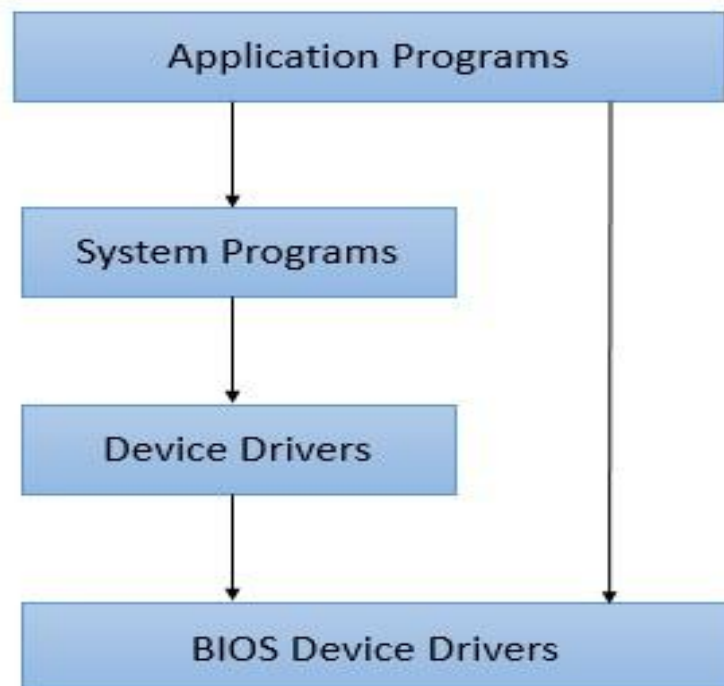


Fig: Simple structure

Advantages of Simple Structure

- Better Performance
- Ease of Development
- Low Overhead
- Faster Debugging (in some cases)
- Ideal for Small Systems

Disadvantages of Simple Structure

- **Lack of Modularity:** No clear separation between functionalities (like file system, memory management, etc.). A change in one part may affect others, making updates and maintenance risky.
- **Poor Maintainability**
- **No Data Hiding**
- **Difficult to Scale:** Cannot be easily extended or adapted for more complex hardware or software requirements.
- **Security Risks**

2. Layered Structure

- In Layered structure, the OS is broken into a number of layers (levels).
- The bottom layer (layer 0) is the hardware, and the topmost layer (layer N) is the user interface.
- These layers are so designed that each layer uses the functions of the lower-level layers.
- The main disadvantage of this structure is that at each layer, the data needs to be modified and passed on, which adds overhead to the system.
- Moreover, careful planning of the layers is necessary, as a layer can use only lower-level layers.
- **UNIX** is an example of this structure.

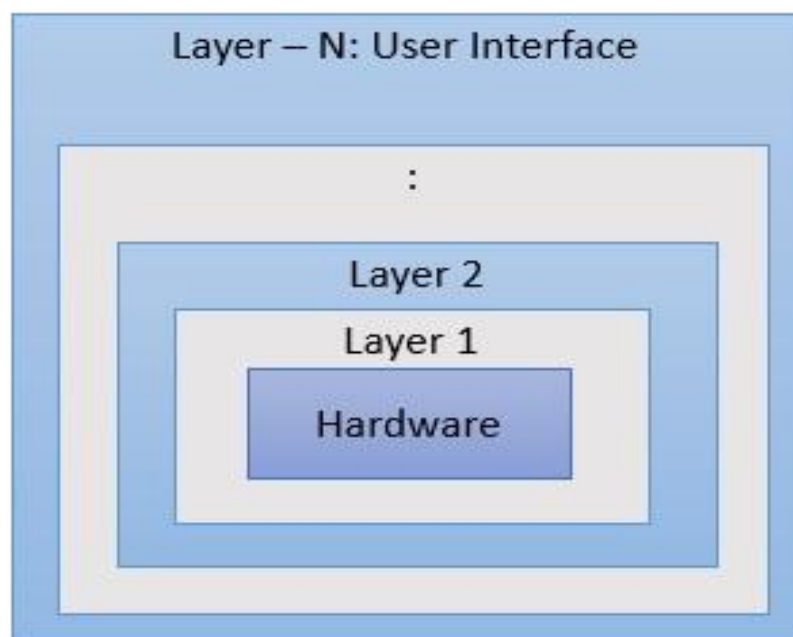


Fig: Layered Structure

Advantages of Layered Structure

- **Modularity:** Each layer is designed to perform a specific function, promoting clean separation of concerns.
- **Ease of Enhancement**
- **Simplified Debugging and Testing**
- **Improved Maintainability**
- **Data Hiding and Encapsulation:** Each layer hides its implementation details from the higher layers. Promotes abstraction, enhancing system security and stability.
- **Better Organization of Code:** Code is logically organized, reducing complexity and duplication.

Disadvantages of Layered Structure

- **Performance Overhead:** The layered approach may lead to extra processing and delays, as each system call must pass through multiple layers.
- **Rigid Structure:** A function in the top layer must rely solely on the lower layer interfaces.
- **Complex Design Requirements:** Designing a layered OS requires careful planning to ensure proper functionality between layers.

3. Modular Structure

- It is considered the best approach for an OS.
- It involves designing a modular kernel.
- **A central kernel is responsible for all major operations of the operating system.** This kernel has a set of core functionality, and other services are loaded as modules dynamically to the kernel at boot time or at runtime.
- For example, **Sun Solaris OS** is an example of a Modular structured operating system.

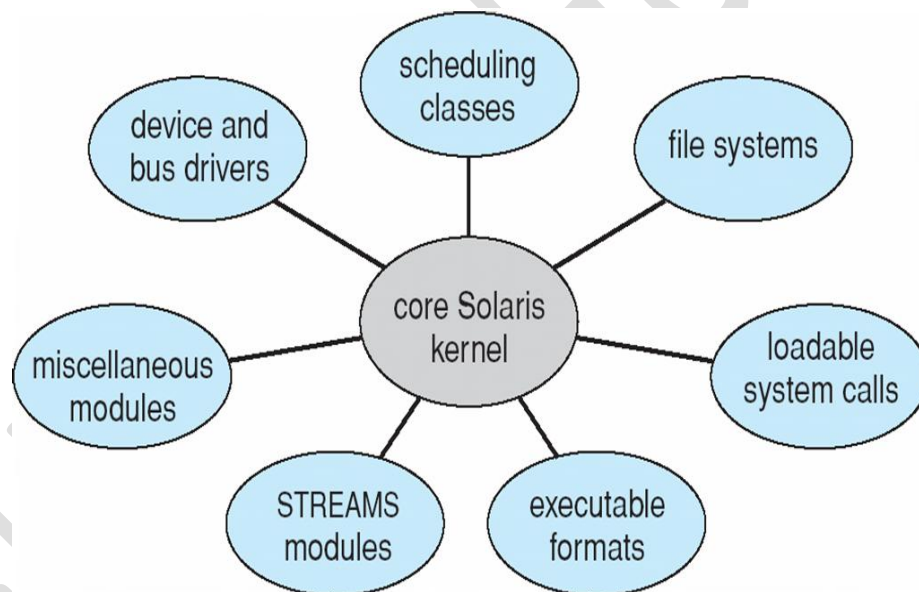


Fig: Solaris Modular Approach

4. Virtual Machines (VMs) Structure

- In this kind of structure, hardware like CPU, memory, and hard disks are abstracted into virtual machines.
- The user can use them and actually configure them using execution contexts.
- Virtual machine takes a good amount of disk space and is to be provisioned.
- Multiple virtual machines can be created on a single physical machine.
- An operating system enables us to run multiple processes concurrently while making it appear as though each one is using a different processor and virtual memory by using CPU scheduling and virtual memory techniques.

Advantages of Virtual Machines (VMs)

- **Highly Customizable**
- **Efficient Resource Utilization**
- **Isolation and Security**
- **Flexibility and Portability**

Disadvantages of Virtual Machines (VMs)

- **Performance Overhead**
- **Complex Designing**
- **Security Risks in Misconfigured Systems**

5. Monolithic Structure

- A monolithic structure is a type of operating system architecture where the **entire operating system is implemented as a single large process in kernel mode.**
- Essential operating system services, such as process management, memory management, file systems, and device drivers, are combined into a single code block
- **MS-DOS or early UNIX.**

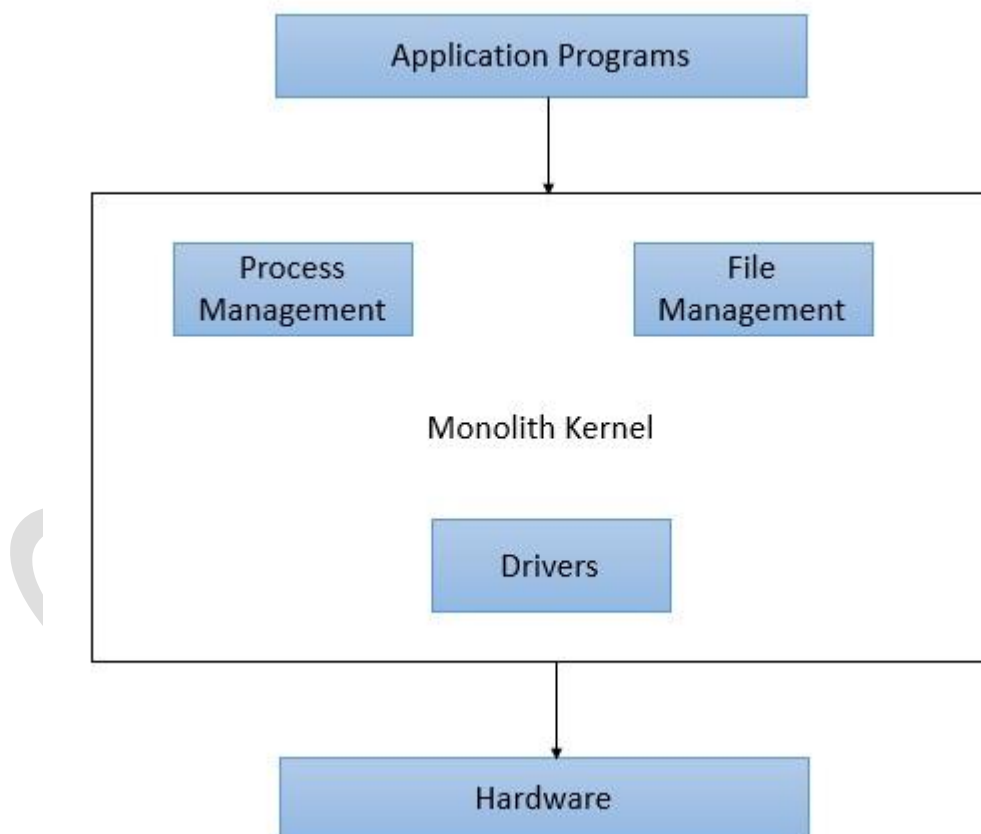


Fig: Monolithic Kernel

6. Micro-Kernel Structure

- Micro-Kernel structure designs the operating system by **removing all non-essential components from the kernel and implementing them as system and user programs.**
- *This results in a smaller kernel called the micro-kernel.*
- **Advantages of this structure are that all new services need to be added to user space and do not require the kernel to be modified.**
- Thus, it is more secure and reliable, as if a service fails, then the rest of the operating system remains untouched.
- **Mac OS** is an example of this type of OS.

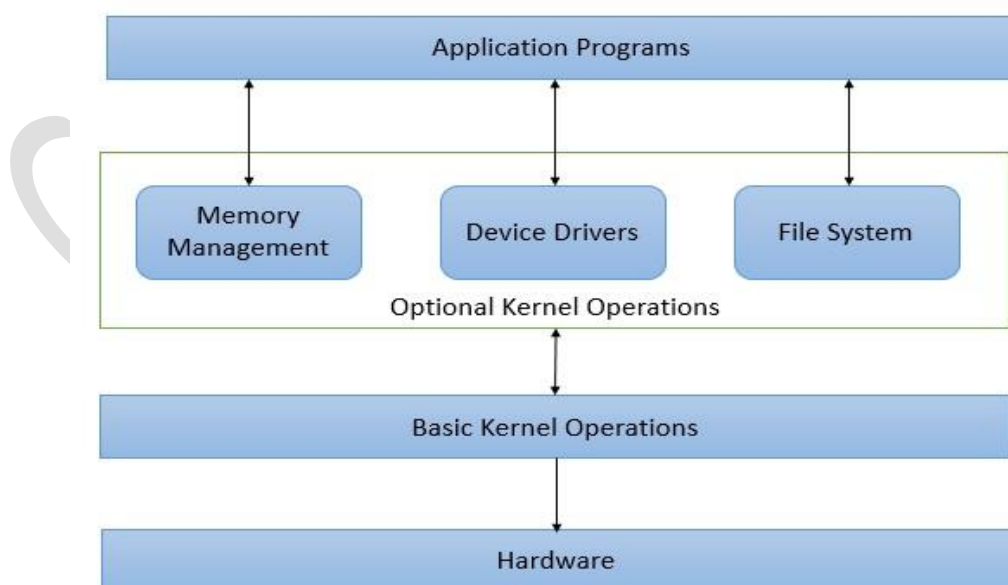


Fig: Micro Kernel

7. Hybrid-Kernel Structure

- A hybrid kernel structure is nothing but just a combination of both a monolithic kernel structure and a microkernel structure.
- Basically, it combines properties of both monolithic and micro-kernel and makes a more advanced and helpful approach.
- It implements speed and design of monolithic and modular, and stability of micro-kernel structure.

What types of Structure do Windows & Linux have?

- *Linux – Monolithic with Modular features*
- *Windows NT and later (including Windows 10/11) – Hybrid Kernel*

8. Exo-Kernel Structure

- Exo-Kernal, a structured operating system, was designed and developed at MIT.
- The aim of this design was to keep Kernel size minimal while allowing the application programs to manage hardware resources directly.
- The purpose of removing the abstraction of the operating system for hardware resources was to enable

application programmers to write high-performance code while the Exo-kernel handles other operations.

Advantages

- High Performance
- Application Control

Disadvantages

- Unreliable and Unsafe
- Complex Design